



**VASANTDADA PATIL PRATISHTHAN'S
COLLEGE OF ENGINEERING AND VISUAL ARTS**

TRUST REGD. NO. F 6736 (MUMBAI)

NAAC ACCREDITATION FOR 5 YEARS : (26.03.2019 to 27.03.2024)

NBA ACCREDITED COURSES : (During 15.03.2012 to 14.03.2015) COMP. ENGG./ ELEX. ENGG. / EXTC. ENGG./IT.
(F.No 28-110 / 2010 - NBA Dated 02/04/2012)

Department of AI &DS /COMP/ IT Engineering

IA-I Examination SH-2023

Subject: Ethical Hacking

Class/Sem: TE/V

Time: 3:30 to 04:30

Date: 12/09/2023

Max. Marks: 20

N.B.:

1. **Question No. 1** is compulsory
2. Answer any **three** out of remaining **four** questions.
3. Each question carry **05** marks.
4. Assume suitable data wherever required but justify the same.

Que No	Questions	Marks	CO	BL	PI
1	a. Explain different steps involved in Ethical Hacking?	2	CO1	L1	1.4.1
	b. Explain biometric authentication & its application?	2	CO3	L1	1.4.1
	c. List out various attacks done by hackers on system?	1	CO1	L2	1.4.1
2	Elaborate in detail public key & private key with encryption with proper diagram?	5	CO3	L4	1.4.1
3	Determine the terms reconnaissance, scanning, vulnerability assessment in detail?	5	CO2	L3	1.4.1
4	Interpret the fundamentals of computer networks/IP protocol stack?	5	CO1	L3	3.2.1
5	Illustrate in detail password cracking, penetration testing, social engineering attacks, malware threats?	5	CO2	L2	1.4.1



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(F.No 28-110 / 2010 - NBA Dated 02/04/2012)

Department of Computer Engineering

IA-I Examination SH-2023

Subject : TCS

Time: 11:00 to 12:00

Date: 8/9/23

Class/Sem : TE-A & B/ V

Max. Marks : 20

1. Question No.1 is compulsory.
2. Answer any three out of remaining four question.
3. Each question carry **05** marks.
4. Assume suitable data wherever required but justify the same.

Q. No	Questions	Marks	CO	BL	PI																											
1	a) Write a short note on Mealy Machine	2	CO1	1	1.3.1																											
	b) Draw the DFA for language over{a,b}: all string starting with abb	2	CO1	3	1.4.1																											
	c) Define Regular Expression.	1	CO2	1	1.1.1																											
2	Construct a NFA to accept the language represented by $a^*b^*c^*$. Convert the NFA to its equivalent DFA	5	CO1	3	2.1.1																											
3	Difference between NFA & DFA	5	CO1	2	2.3.2																											
4	Explain With an example Chomsky Hierarchy.	5	CO2	2	1.1.1																											
5	Construct the minimum state automata equivalent to given DFA <table><tr><td></td><td>0</td><td>1</td></tr><tr><td>→q0</td><td>q1</td><td>q0</td></tr><tr><td>q1</td><td>q0</td><td>q2</td></tr><tr><td>q2</td><td>q3</td><td>q1</td></tr><tr><td>q3*</td><td>q3</td><td>q0</td></tr><tr><td>q4</td><td>q3</td><td>q5</td></tr><tr><td>q5</td><td>q6</td><td>q4</td></tr><tr><td>q6</td><td>q5</td><td>q6</td></tr><tr><td>q7</td><td>q6</td><td>q3</td></tr></table>		0	1	→q0	q1	q0	q1	q0	q2	q2	q3	q1	q3*	q3	q0	q4	q3	q5	q5	q6	q4	q6	q5	q6	q7	q6	q3	5	CO1	3	2.1.1
	0	1																														
→q0	q1	q0																														
q1	q0	q2																														
q2	q3	q1																														
q3*	q3	q0																														
q4	q3	q5																														
q5	q6	q4																														
q6	q5	q6																														
q7	q6	q3																														

**Department of Computer Engineering**
IA1 Examination SH-2023

Subject: Software Engineering
Time: 3:30pm to 4:30pm

Date: 8/09/23

Class/Sem: TE-A, TE-B/V
Max. Marks: 20

N.B.:

1. Question No. 1 is compulsory
2. Answer any **three** out of remaining **four** questions.
3. Each question carry **05** marks.
4. Assume suitable data wherever required but justify the same.

Q. No	Questions	Marks	CO	BL	PI																													
1	a) Discuss the Capability Maturity Model.	2	CO1	BL2	2.6.2																													
	b) Sketch an Activity Node Diagram based on the following Data:	2	CO2	BL3	3.8.1																													
	<table><tr><th>Activity</th><th>Predecessor</th><th>Duration</th></tr><tr><td>A</td><td>-</td><td>3</td></tr><tr><td>B</td><td>-</td><td>4</td></tr><tr><td>C</td><td>-</td><td>7</td></tr><tr><td>D</td><td>A</td><td>3</td></tr><tr><td>E</td><td>A, B</td><td>2</td></tr><tr><td>F</td><td>C</td><td>6</td></tr><tr><td>G</td><td>D, E</td><td>4</td></tr><tr><td>H</td><td>E, F</td><td>2</td></tr><tr><td>I</td><td>G, H</td><td>5</td></tr></table>	Activity	Predecessor	Duration	A	-	3	B	-	4	C	-	7	D	A	3	E	A, B	2	F	C	6	G	D, E	4	H	E, F	2	I	G, H	5			
Activity	Predecessor	Duration																																
A	-	3																																
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H	E, F	2																																
I	G, H	5																																
	c) Differentiate between Organic and Semi-Detached projects in COCOMO Model.	1	CO3	BL2	2.6.4																													
2	Explain any one of the techniques of the Agile Process model with its advantages.	5	CO1	BL2	2.6.4																													
3	Discuss in detail about Scenario Based Modelling. Also sketch a Use Case diagram for Hospital Management System.	5	CO2	BL3	3.6.2																													
4	Solve the function point, for software application by using following given Data: The number of User Inputs(Avg): 22, The number of User Outputs(Low): 45, The number of User Enquiries(High): 06, The number of Internal Logical Files(Avg): 05, The number of External Interface Files(Low): 02, The Complexity Adjustment Factors are average.	5	CO3	BL3	4.7.1																													
5	Sketch a Level 0 and Level 1 DFD for E-Commerce Website.	5	CO2	BL3	3.6.2																													



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Department of Computer Engineering

IA1 Examination SH-2023

Subject: Computer Network
Time: 12 to 1PM

Class/Sem: TE/ V
Date: 11/09/23

Div A & B
Max. Marks: 20

N.B.:

1. Question No. 1 is compulsory
2. Answer any **three** out of remaining **four** questions.
3. Each question carry **05** marks.
4. Assume suitable data wherever required but justify the same.

Q. No	Questions	Marks	CO	BL	PI
1	a) Give the Difference between various transmission media in detail.	2	CO2	4	2.2.5
	b) Compare the OSI reference model & TCP/IP Model.	2	CO1	4	1.4.1
	c) state the name of data units at each layer of OSI model	1	CO1	1	1.4.1
2	Describe types of Aloha Protocols in detail.	5	CO3	2	1.1.1
3	Explain any one type of topology with diagrams, advantages and disadvantages.	5	CO1	2	2.2.4
4	Explain CSMA protocol in detail.	5	CO3	2	2.2.5
5	Apply the CRC error code for the message M= 1110010110 with $G=X^3+1$, construct the polynomial code at the sender's side.	5	CO3	4	5.3.4



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Department of Computer Engineering
Mid Term Examination SH-2023

Class/Sem: TE / V

Max. Marks: 20

Subject: DWM
Time: 3.30pm to 4.30pm

Date: 11/09/2023

N.B.:

1. Question No. 1 is compulsory
2. Answer any **three** out of remaining **four** questions.
3. Each question carry **05** marks.
4. Assume suitable data wherever required but justify the same.

Q. No	Questions	Marks	CO	BL	PI																																												
1	a) The following data are a list of All Electronics prices for commonly sold items. The numbers have been sorted: 1, 1, 5, 5, 5, 5, 8, 8, 10, 10, 10, 10, 12, 14, 14, 14, 15, 15, 15, 15, 15, 15, 18, 18, 18, 18, 18, 18, 18, 18, 20, 20, 20, 20, 20, 20, 20, 21, 21, 21, 21, 25, 25, 25, 25, 25, 28, 28, 30, 30, 30. Draw histogram for given dataset	2	CO2	BL3	2.1.2																																												
	b) Differentiate between Data Mart and Data Warehouse	2	CO1	BL2	1.4.1																																												
	c) List two data mining classification algorithms	1	CO3	BL1	1.4.1																																												
2	Explain different data pre-processing techniques	5	CO2	BL2	1.4.1																																												
3	Use Information Gain (ID3) as attribute selection method & construct Decision Tree <table border="1"><thead><tr><th>Age</th><th>Competition</th><th>Type</th><th>Profit</th></tr></thead><tbody><tr><td>Old</td><td>Yes</td><td>Software</td><td>Down</td></tr><tr><td>Old</td><td>No</td><td>Software</td><td>Down</td></tr><tr><td>Old</td><td>No</td><td>Hardware</td><td>Down</td></tr><tr><td>Mid</td><td>Yes</td><td>Software</td><td>Down</td></tr><tr><td>Mid</td><td>Yes</td><td>Hardware</td><td>Down</td></tr><tr><td>Mid</td><td>No</td><td>Hardware</td><td>Up</td></tr><tr><td>Mid</td><td>No</td><td>Software</td><td>Up</td></tr><tr><td>New</td><td>Yes</td><td>Software</td><td>Up</td></tr><tr><td>New</td><td>No</td><td>Hardware</td><td>Up</td></tr><tr><td>New</td><td>No</td><td>Software</td><td>Up</td></tr></tbody></table>	Age	Competition	Type	Profit	Old	Yes	Software	Down	Old	No	Software	Down	Old	No	Hardware	Down	Mid	Yes	Software	Down	Mid	Yes	Hardware	Down	Mid	No	Hardware	Up	Mid	No	Software	Up	New	Yes	Software	Up	New	No	Hardware	Up	New	No	Software	Up	5	CO3	BL3	2.1.2
Age	Competition	Type	Profit																																														
Old	Yes	Software	Down																																														
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Mid	No	Software	Up																																														
New	Yes	Software	Up																																														
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4	Model Star Schema for given case study. Consider a data Warehouse for a hospital where there are three dimensions a) Doctor, b) Patient and c) Time. Consider two measures <i>count</i> & <i>fees</i> .	5	CO1	BL3	3.3.1																																												
5	Apply Naive Bayes classification technique to classify given tuple X = (Age = "<= 30", Income = "Medium" Student = "Yes" Credit_rating = "Fair")	5	CO3	BL3	2.1.2																																												



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Department of Computer Engineering

IA1 Examination SH-2023

Subject: IP(Internet Programming)

Class/Sem: TE-A & B /V

Date:12/09/2023

Time: 12:00 To 01:00 PM

Max. Marks: 20

N.B.:

1. Question No. 1 is compulsory
2. Answer any **Three** out of remaining **Four** questions.
3. Each question carry **05** marks.
4. Assume suitable data wherever required but justify the same.

Q. No	Questions	Marks	CO	BL	PI
1	a) Explain CSS Animation properties.	2	CO1	2	1.6.1
	b) Explain Document Object Model.	2	CO2	2	1.7.1
	c) Write a CSS code to demonstrate the use of text-shadow property.	1	CO1	3	2.5.1
2	Illustrate a Javascript code to check Mobile Number. (Mobile numbers should start with 9 or 8 and length is 10 digits.)	5	CO2	3	2.5.1
3	List and Explain Session handling Techniques.	5	CO3	2	1.7.1
4	Define Client, Server? Explain client server communication with suitable diagram.	5	CO1	2	1.7.1
5	Describe Servlet Life Cycle with diagram.	5	CO3	2	1.5.1